

Seriation, Pipestem Dating, Mean Ceramic Dating

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Seriation (and Chronological Inference)

- Seriation is a set of **methods** designed to solve the problem of chronological inference.
- The methods are based **models** of how **variable values** tend to be distributed across **units** at successive points in time.
 - **Units:** things that can be inferred to be the results of events with restricted durations, *e.g.* artifacts, assemblages of artifacts.
 - **Variable values:** the values of variables that characterize the units, *e.g.* *Shapes of artifacts, type frequencies in assemblages*
- Methods specify **procedures** for ordering units along a single dimension, based on the distribution of the values across them.
- The procedures allow us to achieve an order of units that fits the model.
- In a particular case, the ordering of units is a “**chronological hypothesis**”

- **Occurrence Seriation:**

Units: assemblages
Variables: discrete (1/0)

- **Frequency Seriation:**

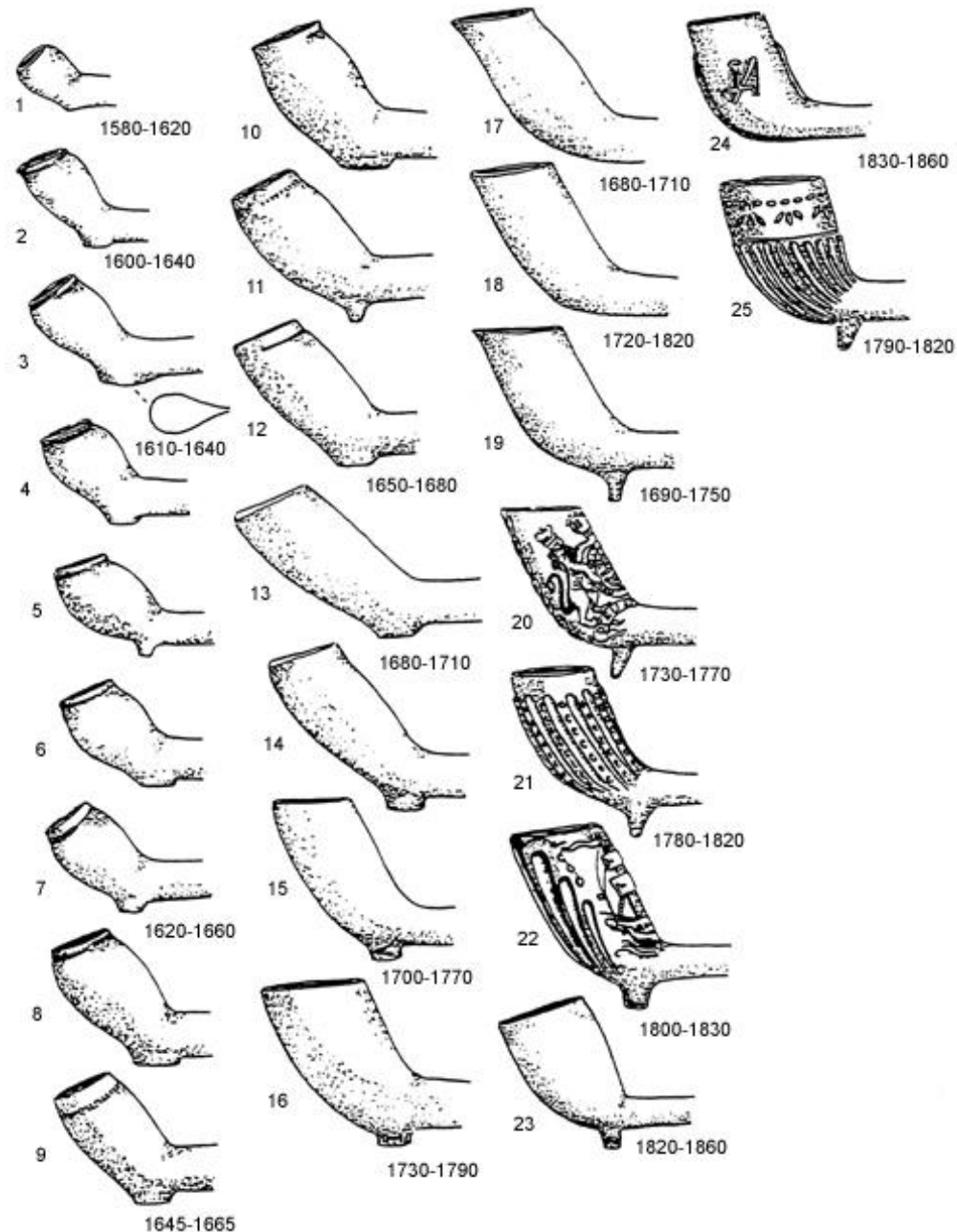
- Units: assemblages
- Variables : continuous (% frequency of “types”)

- **Phyletic Seriation:**

- Units: artifacts or elements
- Variables: discrete or continuous

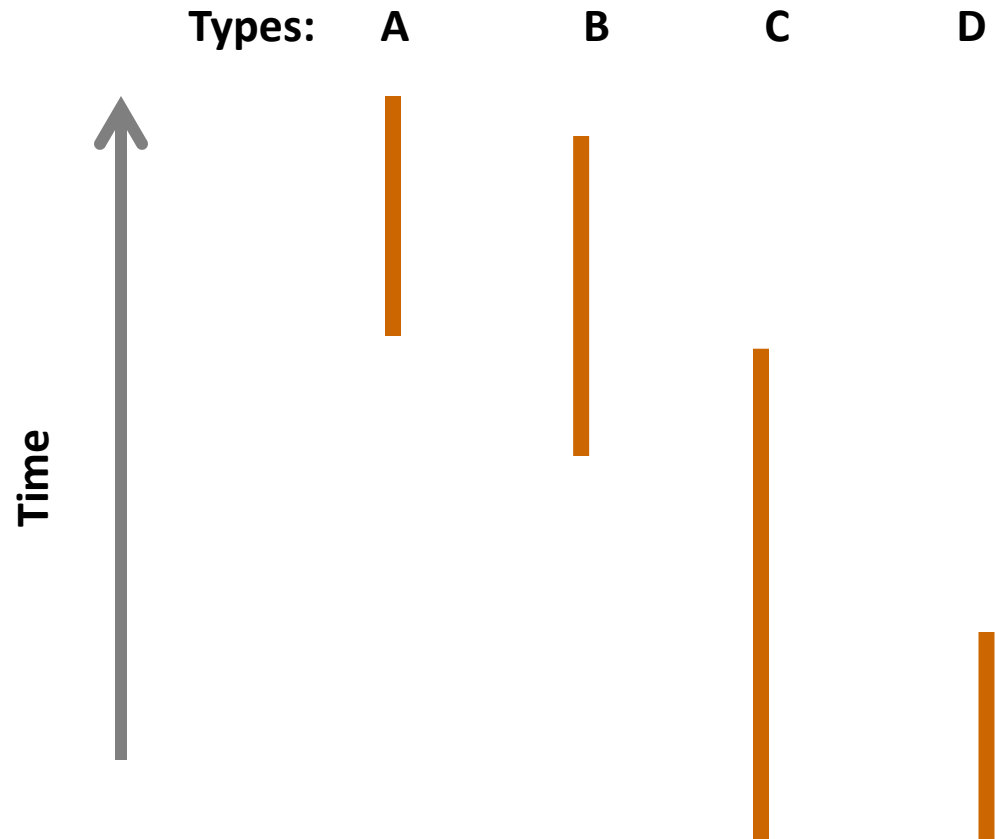
Phyletic Seriation Model

- Objects that are close together in time are more similar than those that are far apart.
- Question: Which variables should we use of measure similarity?



Occurrence Seriation Model

- The temporal distribution of types (characters) is continuous
- There is overlap in the temporal distribution of characters



Occurrence Seriation Method

- Given a set of units, characterized in terms of presence absence of attributes (types), the ordering of units that minimizes the number of absences among the presences is likely to be a chronology.
- “*the concentration principle*”: minimize the number of gaps
- Question: Which variables (types)?

Occurrence Seriation

- Ugly raw data:

<u>Types:</u>		<u>D</u>	<u>B</u>	<u>A</u>	<u>C</u>
Units:	2		1	1	
	5	1			1
	6	1			1
	4		1		1
	3		1	1	
	1			1	

Occurrence Seriation

- Permute rows to minimize total number of gaps down columns.

<u>Types:</u>		<u>D</u>	<u>B</u>	<u>A</u>	<u>C</u>
Units:	2		1	1	
	5	1			1
	6	1			1
	4		1		1
	3		1	1	
	1			1	

Total Gaps = 5

Occurrence Seriation

- Permute rows to minimize total number of gaps down columns.

<u>Types:</u>		<u>D</u>	<u>B</u>	<u>A</u>	<u>C</u>
Units:	1			1	
	2		1	1	
	3		1	1	
	4		1		1
	5	1			1
	6	1			1

Total Gaps = 0

Occurrence Seriation

- Permute columns to minimize number of gaps across rows.

<u>Types:</u>		<u>D</u>	<u>B</u>	<u>A</u>	<u>C</u>
Units:	1			1	
	2		1	1	
	3		1	1	
	4		1		1
	5	1			1
	6	1			1

Total Gaps = ?

Occurrence Seriation

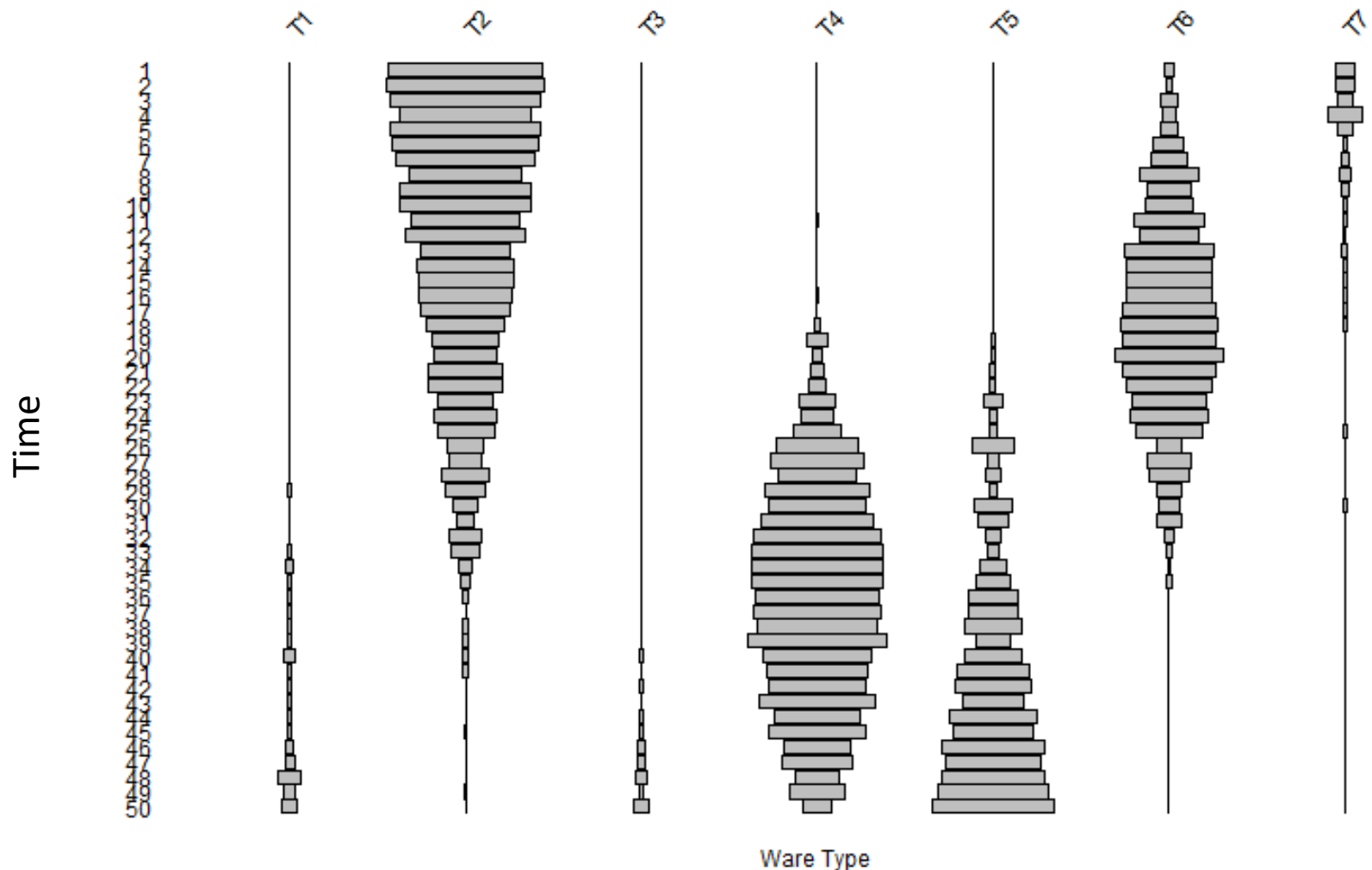
- Permute columns to minimize number of gaps across rows.

<u>Types:</u>		<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
Units:	1	1			
	2	1	1		
	3	1	1		
	4		1	1	
	5			1	1
	6			1	1

Total Gaps = 0

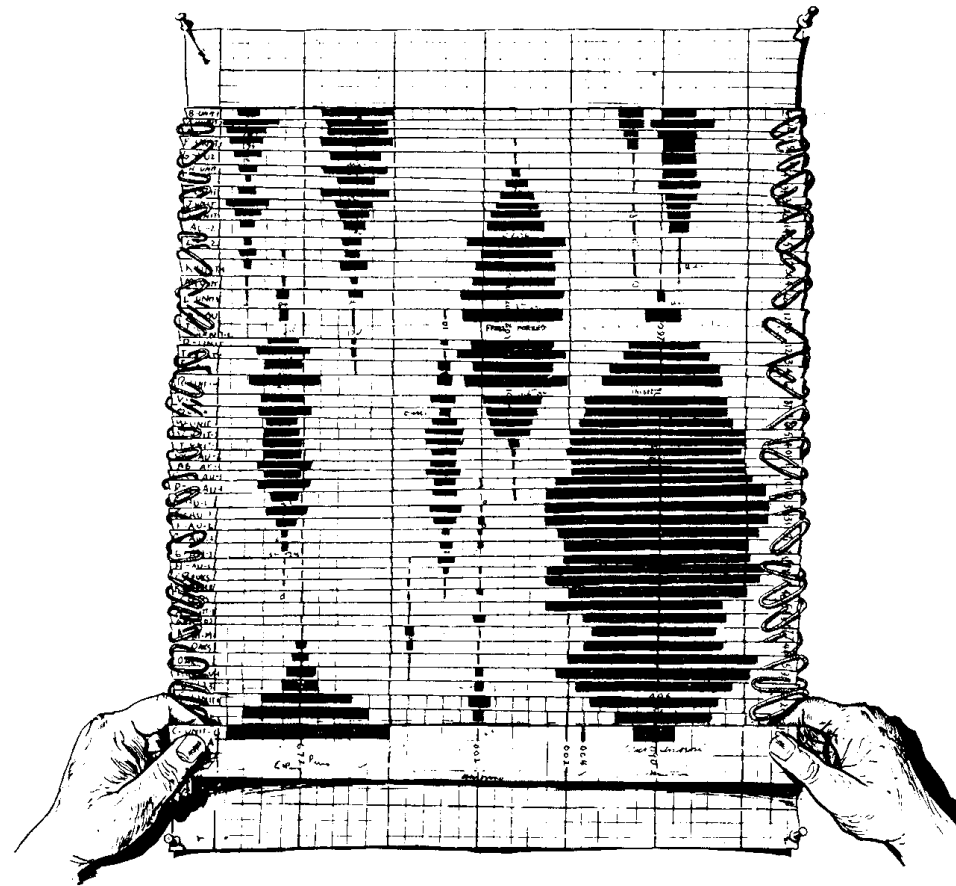
Frequency Seriation Model

- When the distributions of characters (types) are described in relative frequency terms, their shape in time tends to be lenticular or battleship-shaped.



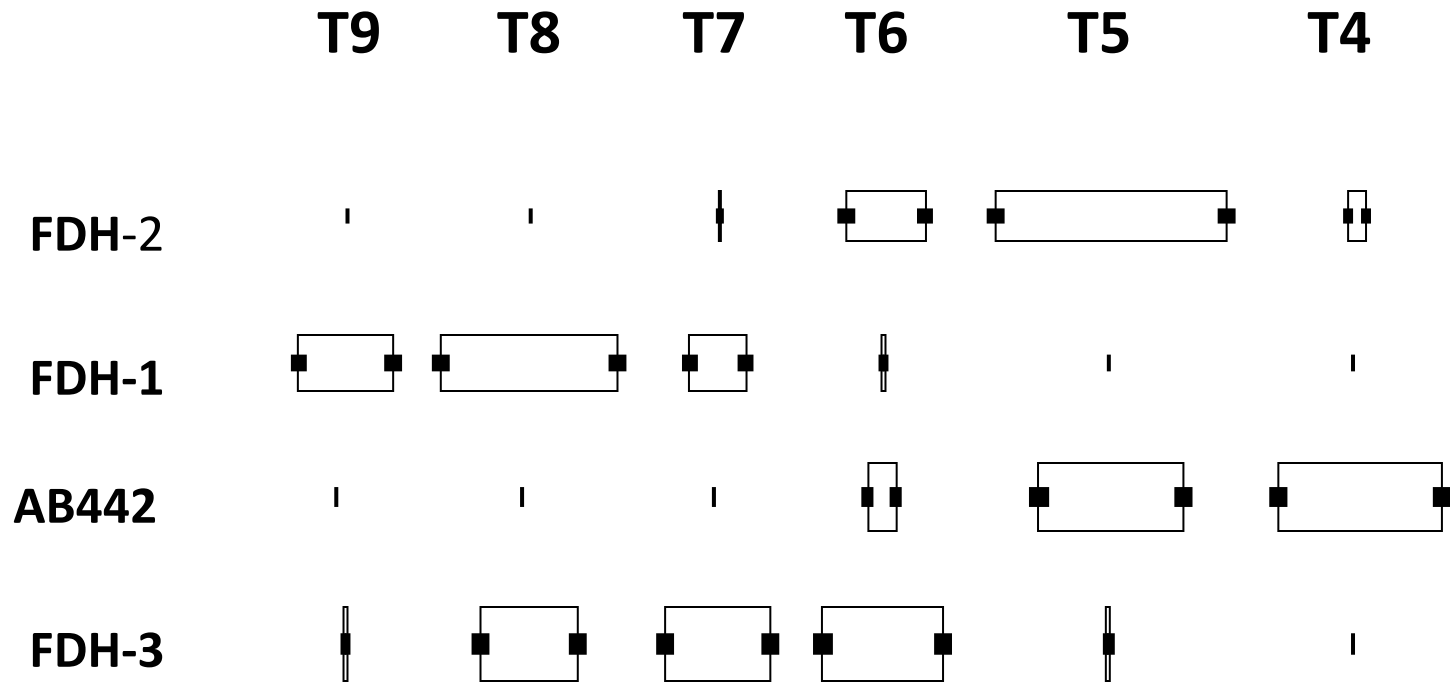
Frequency Seriation Method

- The ordering of units in which battleship-shaped distributions of types emerge is likely to be a chronology.
- Question: Which variables (types)?



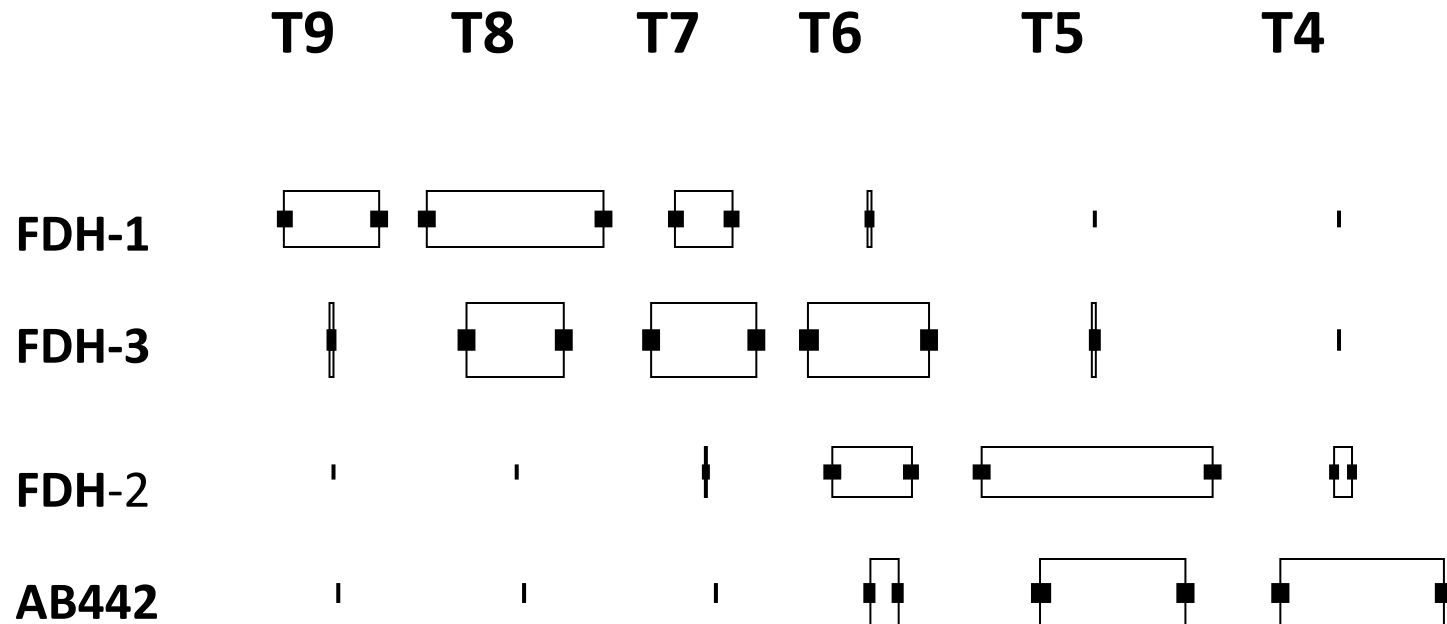
Frequency Seriation

- The ugly raw data



Frequency Seriation

- Permute the rows so that the battleship curves emerge.



Questions

- **What is being dated?**
- **Which way is up?**
- **How can we check the order really is a chronology?**

Questions

- **What is being dated?**

Archaeological assemblages are “*time averaged*”: they integrate artifact breakage and discard events taking place over a period of time. So the event being data is a function of the entire period over which the assemblage accumulated.

cf. Sediments in a deposit.

- **Which way is up?**

We need independent evidence.

- **How can we check the order really is a chronology?**

We need independent evidence.

Conditions

1. Units have similar durations

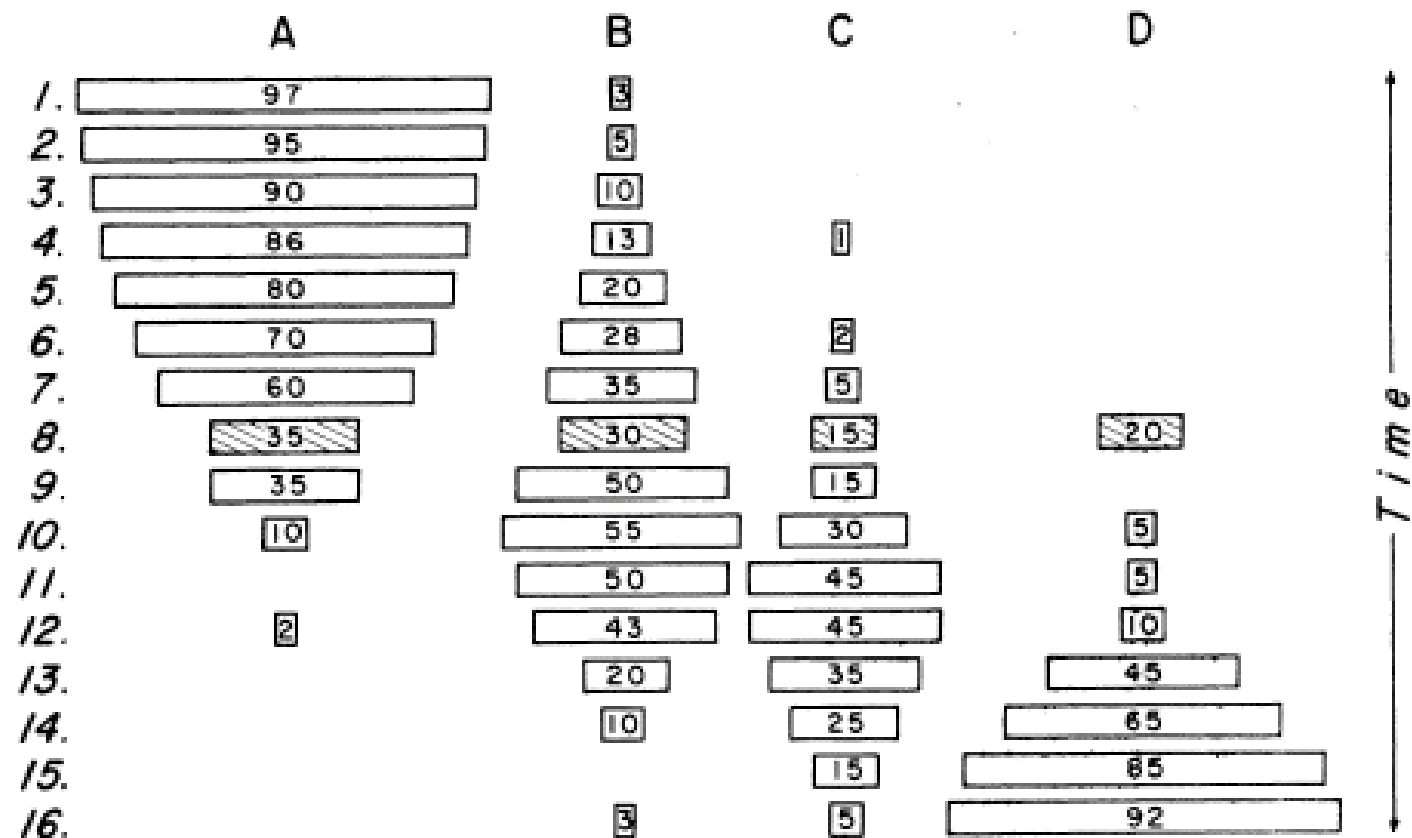


Fig. 3. The effect upon the seriation pattern of one group or unit not of comparable duration with the remainder of the included units. A-D are the classes, 1-16 are the groups of which B is the non-comparable unit.

Conditions

2. Units come from the same “cultural tradition”

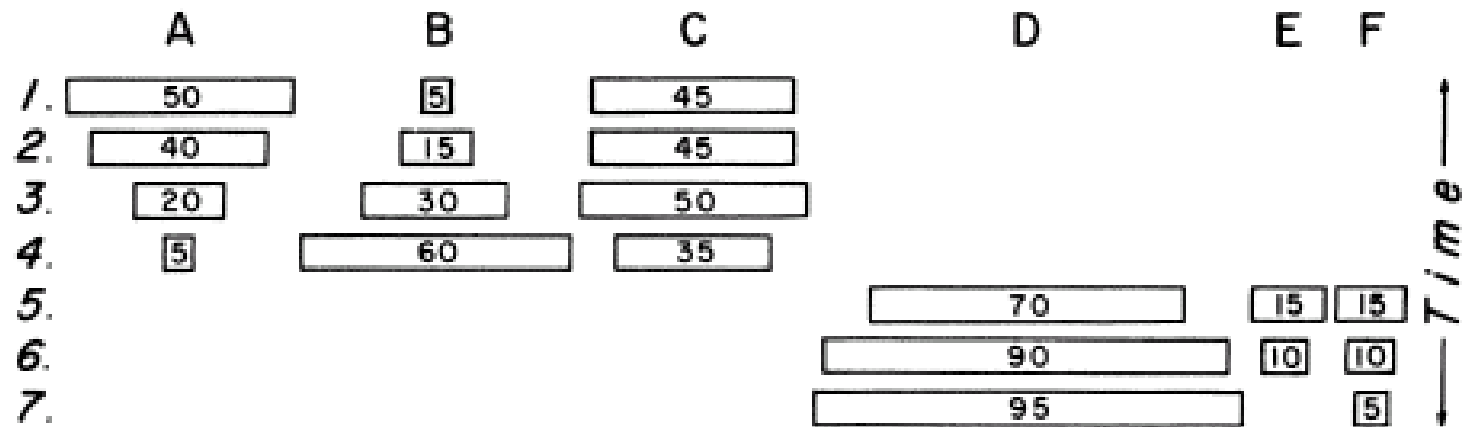


Fig. 4. Two independent orders resulting from the seriation of groups drawn from two different stylistic traditions. A - F are the classes and 1 - 7 are the groups. The placement of 5 - 7 at the bottom of the chart is arbitrary and they could have been placed at the top of the chart with equal justification.

3. Units come from the same “local area”.

“Same Local Area”

- *Deetz and Dethlefsen* show that different local areas (town cemeteries) show different patterns of battleship-shaped curves for grave-stone types.

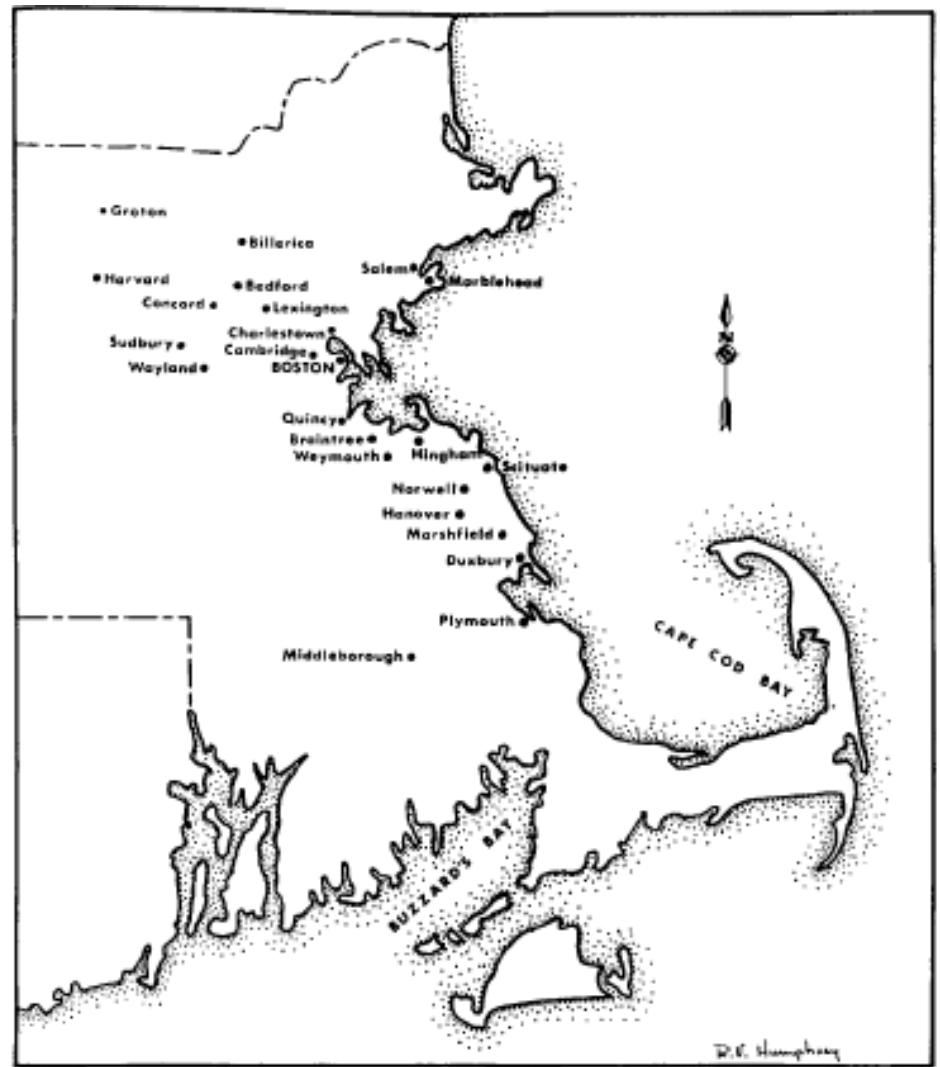


FIG. 1. Eastern Massachusetts, showing cemetery locations.

Source: Dethlefsen, E. and J. Deetz
1966 Death's heads, cherubs and Willow trees: experimental archaeology in colonial cemeteries. *American Antiquity*
31:502-510.

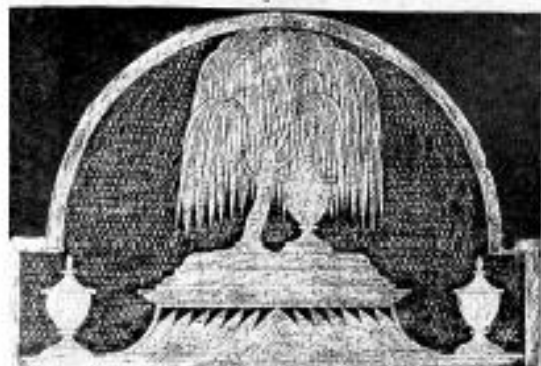
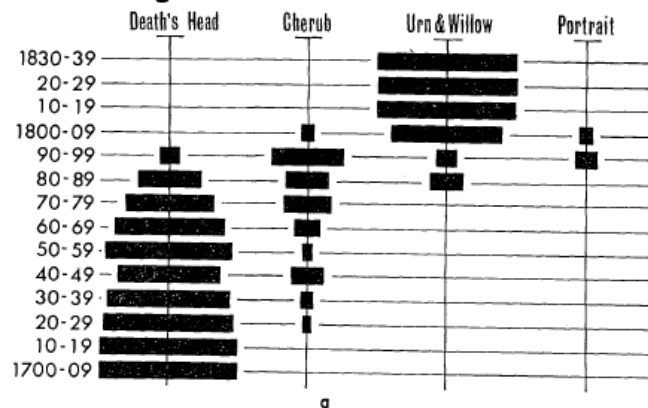
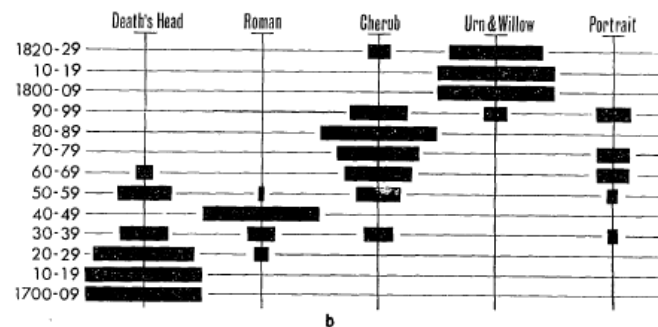


FIG. 2. Universal motifs. a, death's head; b, cherub;
c, urn and willow.

Cambridge



Concord



Plymouth

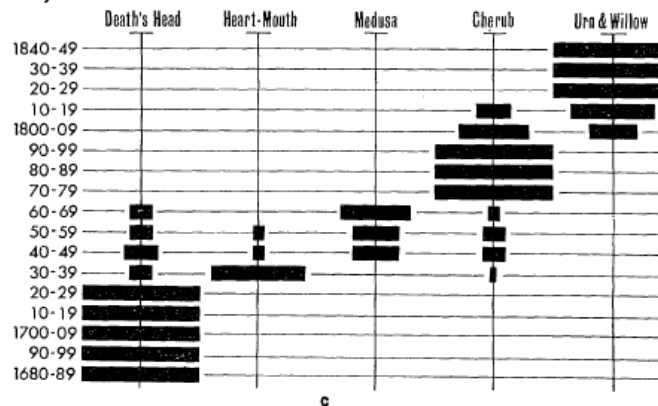


FIG. 3. Graphs showing stylistic sequences in three cemeteries.

“Same Local Area”

- Lipo *et al.* show that you can use the “local area” criterion to identify local areas – neighborhoods within which assemblages show the battleship-shaped curves for ceramic types – a single evolving tradition.

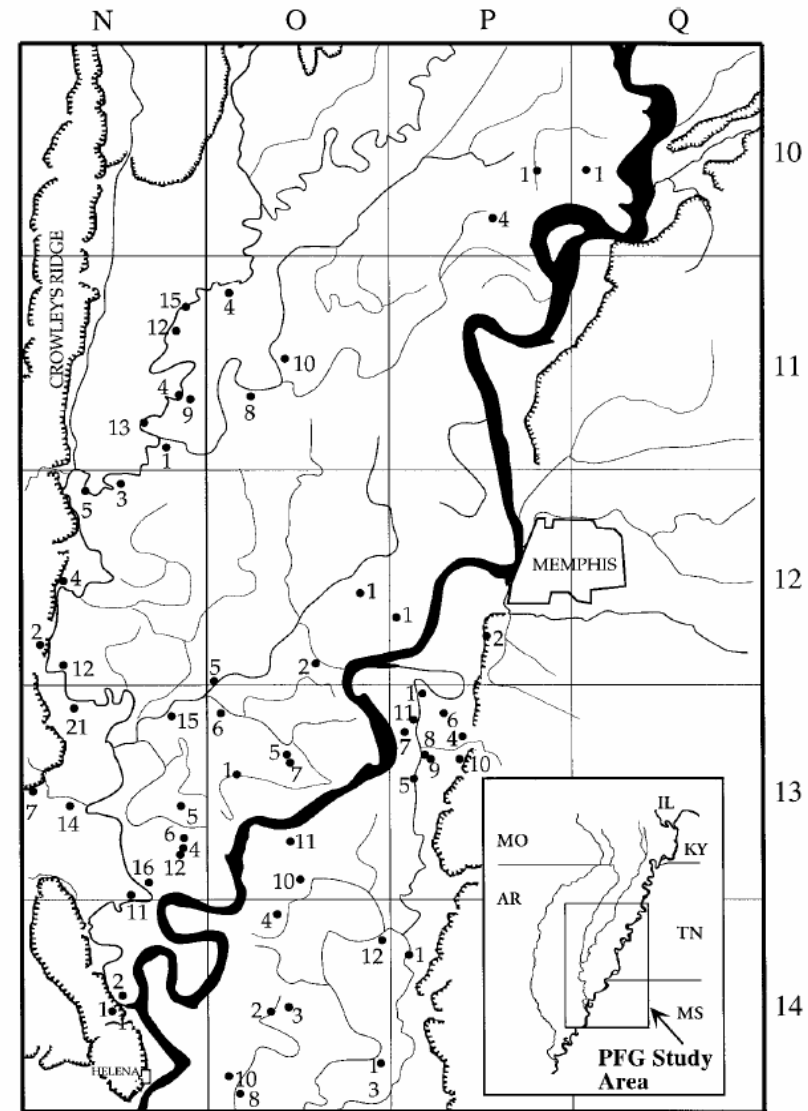


FIG. 9. Map of Lower Mississippi Valley Survey: Assemblages from the St. Francis and Mer regions from Phillips *et al.* (1951).

Source: Lipo, Carl P., Mark E. Madsen, Robert C. Dunnell, and Tim Hunt.
 1997 Population structure, cultural transmission, and frequency seriation. *Journal of Anthropological Archaeology* 16 (4): 301-333.

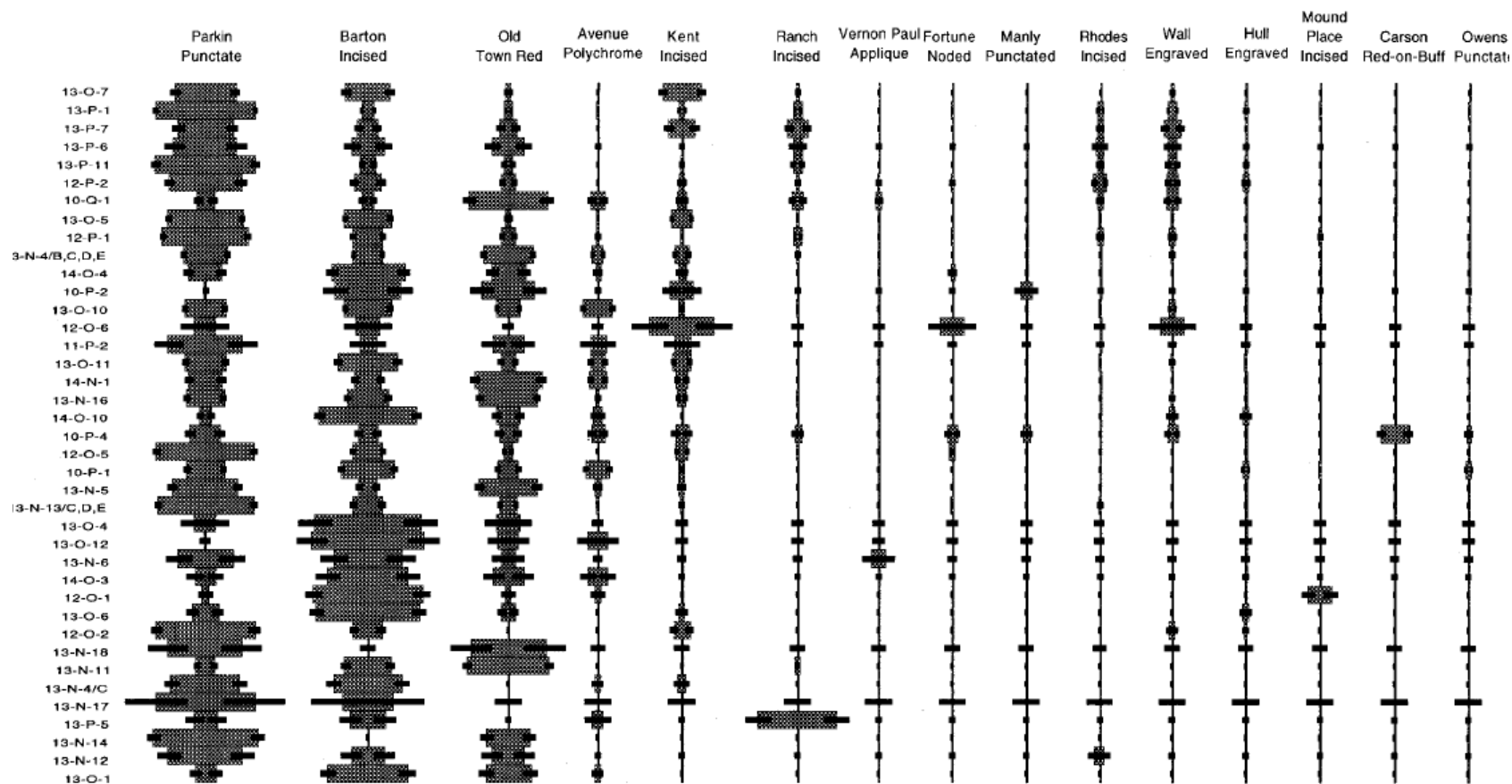


FIG. 11. Ford's seriation of the Memphis area data using only assemblages collected from the surface and decorated, shell tempered ceramic types. It is clear, that the neither the seriations in Fig. 10 nor Fig. 11 meet the expectations of the model.

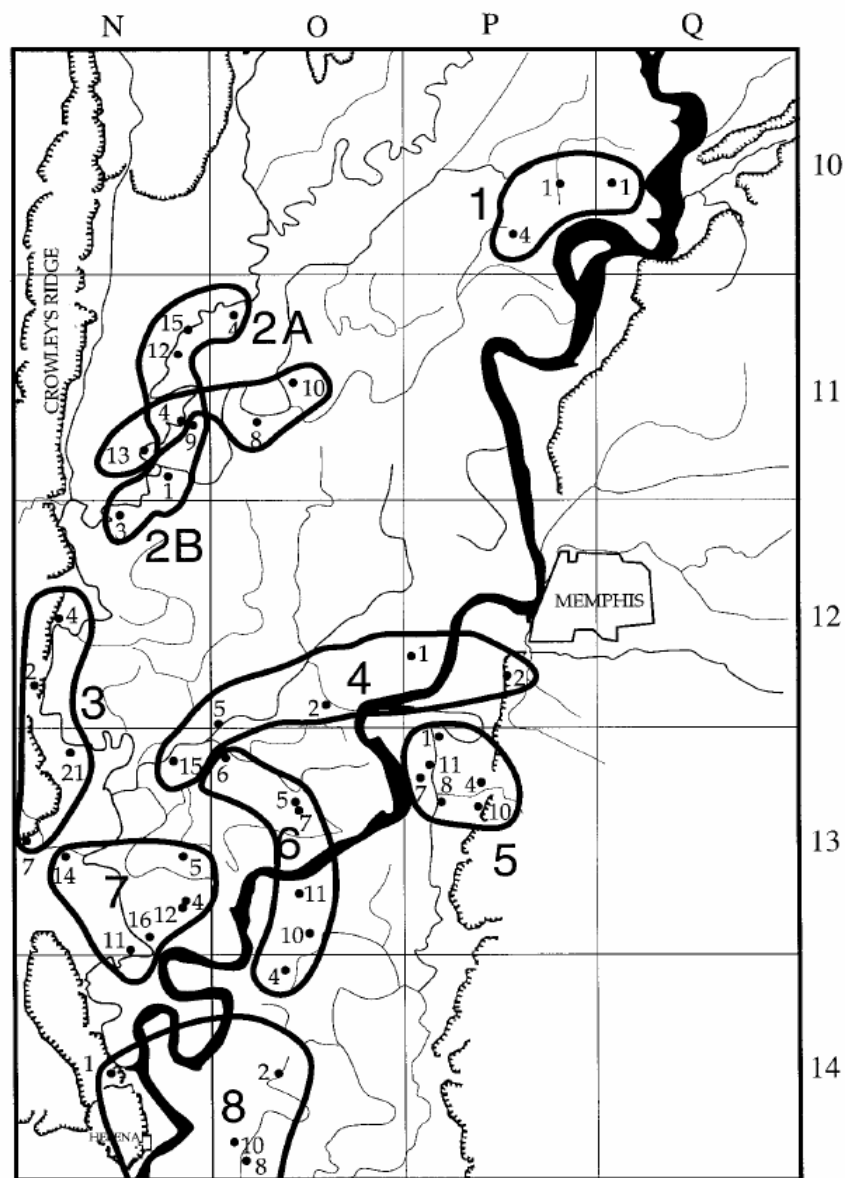


FIG. 13. Geographic distribution of solution clusters of all PFG assemblages. Numbers (1-8) refer to the seriation solutions from Fig. 12.

	Parkin Punctate	Barton Incised	Old Town Red	Avenue Polychrome	Kent Incised	Ranch Incised	Fortune Noded	Vernon Paul Applique	Manly Punctate	Rhodes Incised	Wall Engraved	Hull Engraved	Mound Place Incised	Carson Red-on-Buff	Owens Punctate
Group 1															
10-P-1															
10-P-4															
10-Q-1															
Group 2A															
11-N-4															
11-N-13															
11-O-10															
11-O-8															
Group 2B															
11-O-4															
11-N-15															
11-N-9															
11-N-1															
12-N-3/A&B															
11-N-12															
Group 3															
12-N-2															
13-N-21															
13-N-7															
12-N-4															
13-P-10															
Group 4															
12-O-2															
12-O-5															
13-N-15/C,D,E															
12-P-1															
12-P-2															
Group 5															
13-P-11															
13-P-1															
13-P-8															
13-P-4															
13-P-7															
Group 6															
13-O-5															
13-O-7															
13-O-11															
13-O-10															
14-O-4															
13-O-6															
Group 7															
13-N-14															
13-N-5															
13-N-4/B,C,D,E															
13-N-16															
13-N-11															
Group 8															
16-N-2/B-1															
14-N-1															
14-O-2															
16-N-6															
15-N-6															
14-O-10															
14-O-8															

Conditions

4. No sampling issues.

Assemblages are large enough so that estimates of type proportions in them mirror actual type frequencies in the underlying population of sherds:
sampling error does not obscure patterns.

5. Battleship-Shaped Curves Rule.

The underlying type frequencies really do have Battleship-Shaped curves.
They are “historical types”

Frequency Seriation

Conditions

- Battleship curves ***if***
 - assemblages have similar durations.
 - assemblages from same local area and cultural tradition.
 - little sampling error
 - "historical types" = battleship-shaped curves

Evaluation

- Goodness of fit to the model
 - Do the assemblages and types you have chosen seriate?
 - Evidence of discontinuity in the sequence?
 - Lack of fit is a learning opportunity!
- Agreement with independently derived chronological hypotheses
 - Does the order from one seriation match the order from a second seriation , based on independent data?
 - Does the order correlate with orders based on independent data and independent models theoretical models? (e.g. 14C, dendrochronology, OSL, stratigraphy?)

Type Fossils

- A second approach to chronological inference, based on artifacts.
- Assign units to periods of time, based on the presence/absence of types.
- *Type dates must be known on other grounds*

Type Fossils

Creamware: 1760-1820

Pearlware: 1780-1830



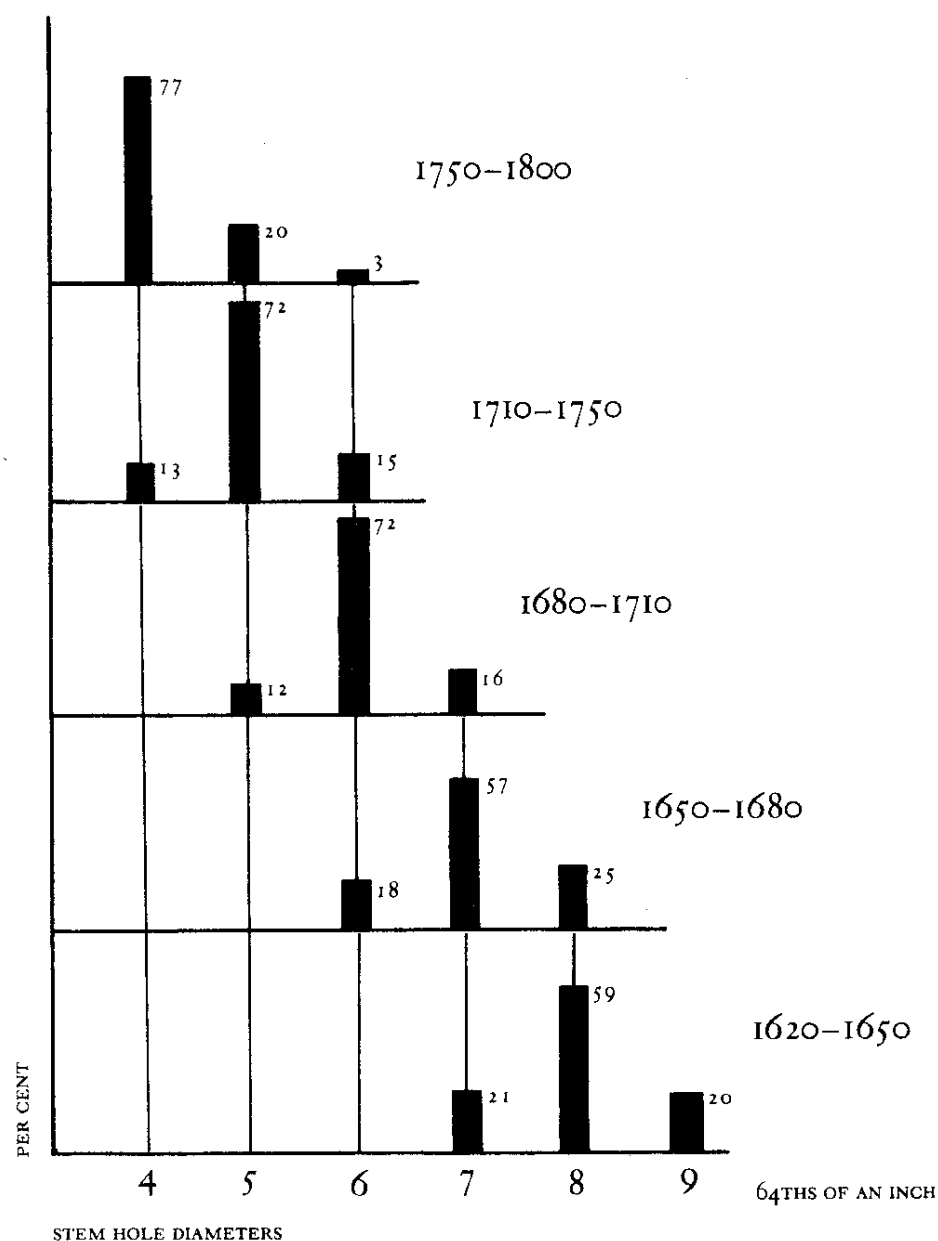
- *Terminus post quem* (TPQ): Date after which
- *Terminus ante quem* (TAQ): Date before which

Hybrid Methods

- Pipestem dating
- Mean ceramic dating
used in historical archaeology (and the Southwestern US)

Pipestem Dating

Harrington histograms



Source:

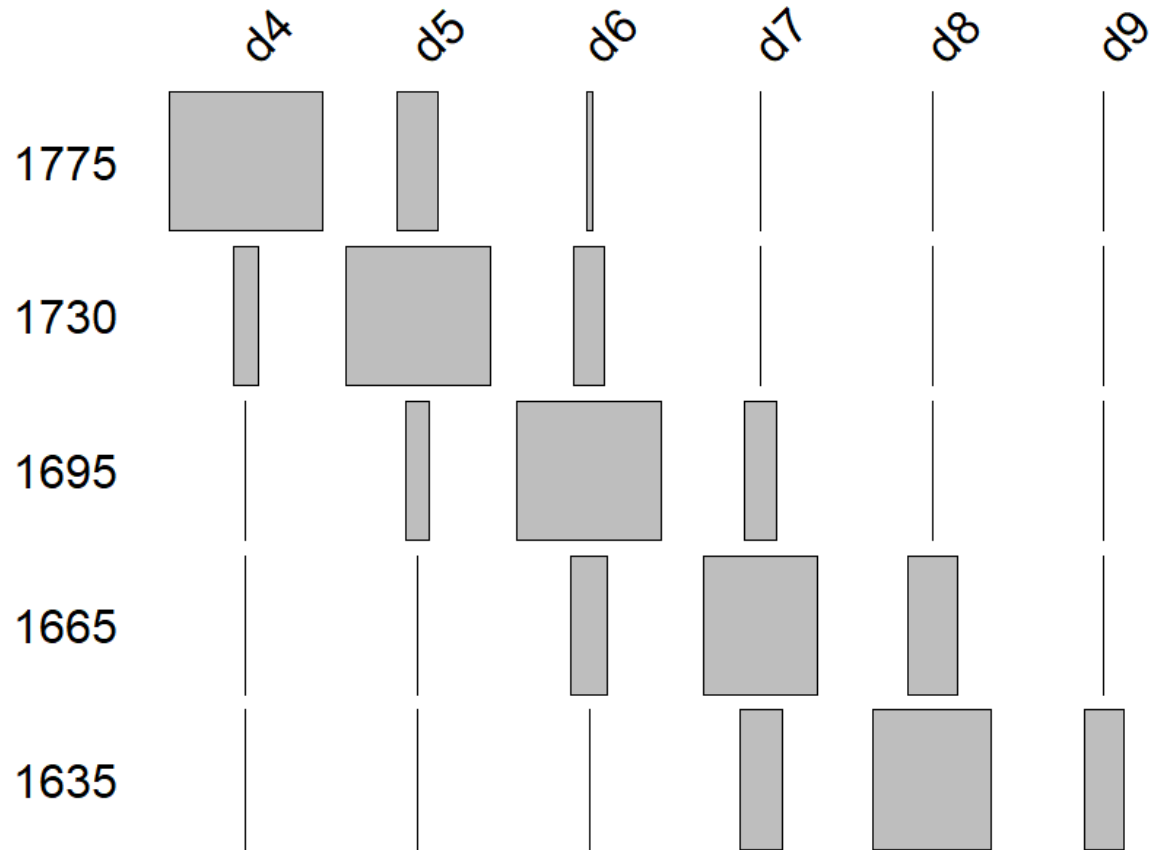
Harrington, JC

1954

Dating stem fragments of 17th and 18th century tobacco pipes. *Quarterly Bulletin of the Archaeological Society of Virginia*

Pipestem Dating

Harrington histograms ... transposed!



Harrington Histograms → Mean Bore Diameters

$$\bar{x} = \frac{\sum_{i=1}^n f_i i}{\sum_{i=1}^n f_i}$$

1. The Data

Median Date	4	5	6	7	8	9	Total
1775	77	20	3	0	0	0	100
1730	13	72	15	0	0	0	100
1695	0	12	72	16	0	0	100
1665	0	0	18	57	25	0	100
1635	0	0	0	21	59	20	100

2. Products

1775	308	100	18	0	0	0
1730	52	360	90	0	0	0
1695	0	60	432	112	0	0
1665	0	0	108	399	200	0
1635	0	0	0	147	472	180

3. Sum of Products

1775	426
1730	502
1695	604
1665	707
1635	799

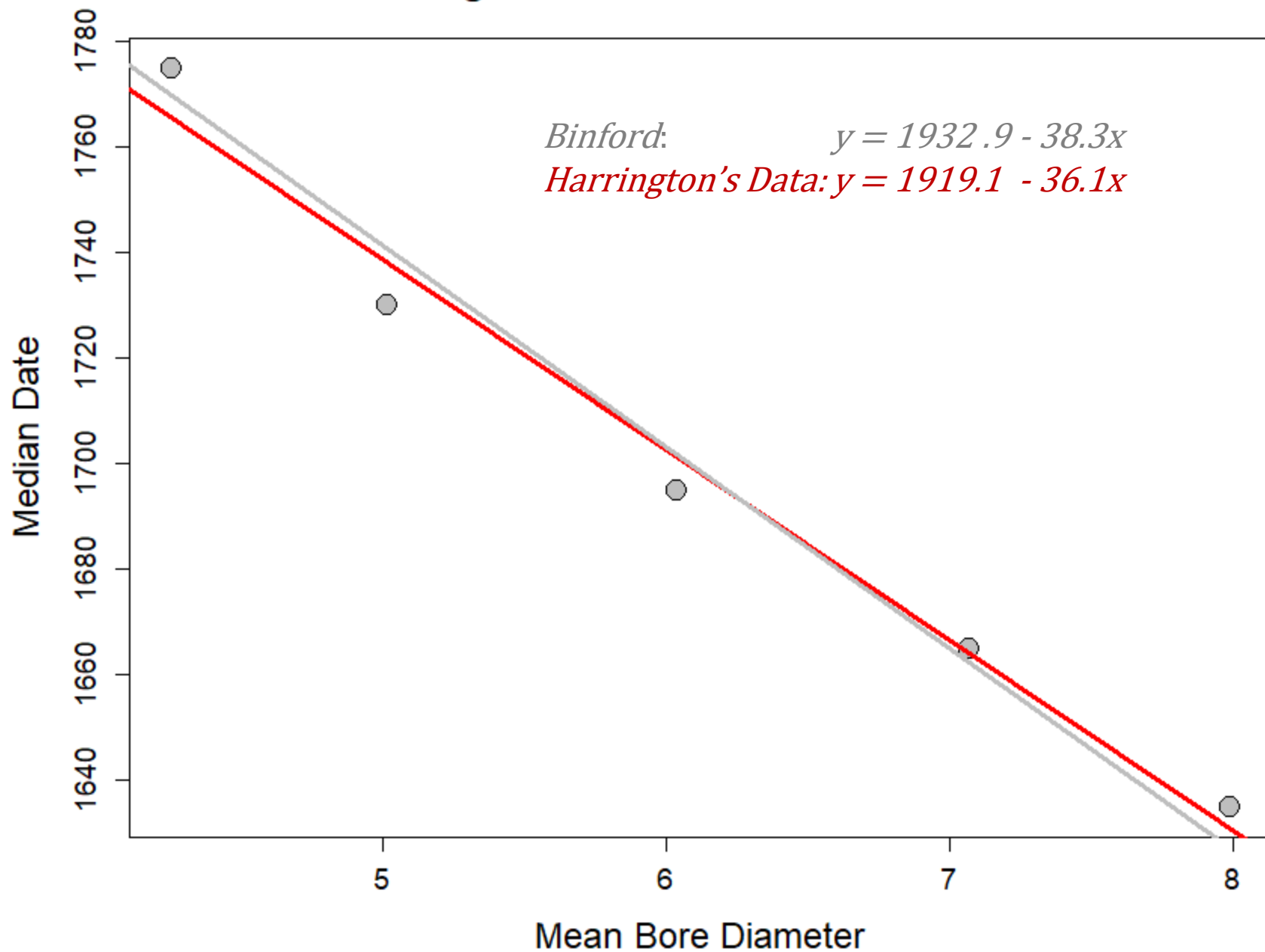
4. Mean Bore Diameter

1775	4.26
1730	5.02
1695	6.04
1665	7.07
1635	7.99

$i = 1$ through n indexes the bore diameter
class values (e.g. 2,3,4,5,6,.../64th inch)

f_i = the frequency (count) for the i 'th class

Regressions of Date on Diameter



Mean Ceramic Dating

$$\bar{x} = \frac{\sum_{i=1}^n f_i m_i}{\sum_{i=1}^n f_i}$$

$i = 1$ through n indexes the ware type classes.

f_i = the frequency (count) for the i 'th ware type.

m_i = manufacturing midpoint for the i 'th ware type

Source:

South, Stanley

1972

Evolution and Horizon as Revealed in Ceramic Analysis in Historical Archaeology. *Conference on Historic Site Archaeology Papers* 6:71-116.

1. The Data

Type:	Creamware	Delft	Pearlware	WhiteSaltGlaze	Total
Median Date:	1790	1750	1805		1760
Dry Well	666	0	0	169	835
Site7	331	90	51	46	518
Bldg.O	492	40	89	36	657
House	102	8	75	2	187
Bldg.T	326	21	156	26	529

2. Products

Dry Well	1192140	0	0	297440
Site7	592490	157500	92055	80960
Bldg.O	880680	70000	160645	63360
House	182580	14000	135375	3520
Bldg.T	583540	36750	281580	45760

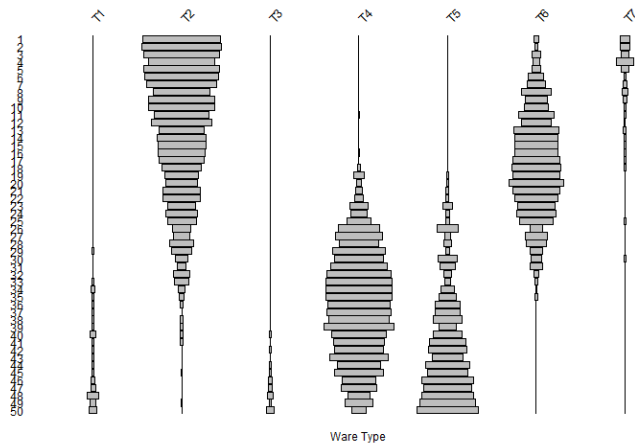
3. Sum of Products

Dry Well	1489580
Site7	923005
Bldg.O	1174685
House	335475
Bldg.T	947630

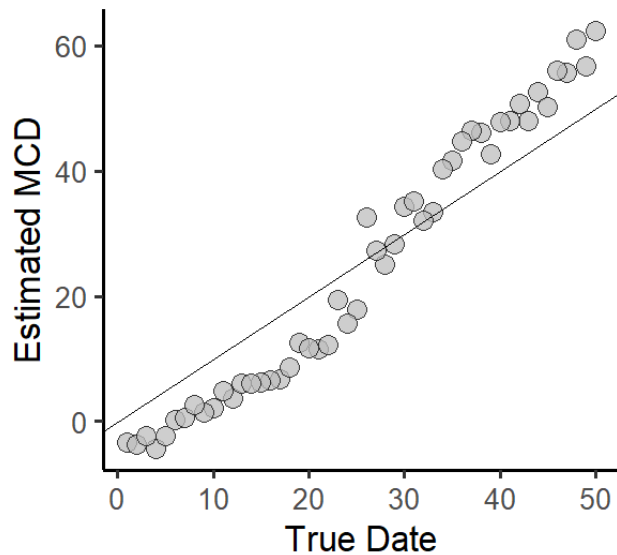
4. MCD

Dry Well	1783.9
Site7	1781.9
Bldg.O	1788.0
House	1794.0
Bldg.T	1791.4

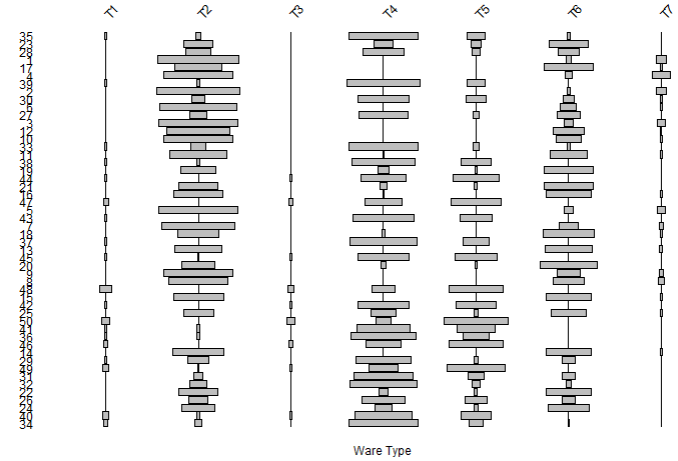
Mean Ceramic Dating Simulation



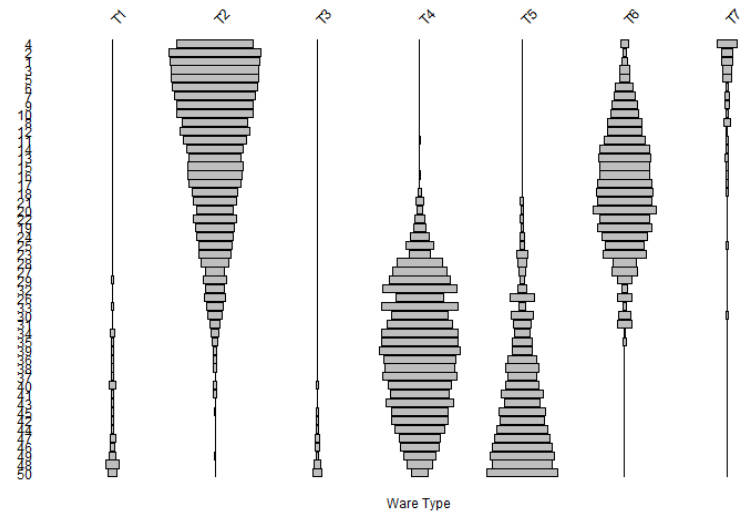
1. Type proportions follow Gaussian curves
Along a temporal gradient ($t = 1, 2, 3 \dots 50$).



3. Estimate MCDs using known type means and
type proportions.



2. Randomize assemblage order



4. Sort assemblages on MCDs and plot



MCDs for Monticello Sites

Median Date	1790	1790	1790	1750	1805	1805	1860	1800	1843	1805	1805	1888	1818	1785	1770	1760
Type	Cr	CrOverg	CrTrans	Delft	Pe	PeAnnul	PeFlow	PeHandp	PeMocha	PePoly	PeShell	PeSpong	PeTrans	PoHandp	PoOverg	StWhite
Dry Well	666	0	0	0	0	0	0	0	0	0	0	0	0	25	197	169
Site7	331	0	0	90	51	7	0	10	0	1	4	0	2	18	1	46
Bldg.O	492	7	2	40	89	11	1	76	2	16	27	0	33	224	124	36
House	102	0	1	8	75	1	2	53	1	11	12	0	54	108	61	2
Bldg.T	326	1	1	21	156	15	0	87	5	75	45	1	113	127	63	26
SmokeH	467	66	0	0	57	2	0	99	28	157	94	0	358	267	202	65
Bldg.R	236	3	1	4	185	14	3	95	4	43	29	8	59	151	48	2
Bldg.L	181	3	0	3	98	4	0	59	3	54	29	0	126	81	59	7
Stewart	286	3	15	0	135	15	0	86	0	340	47	0	1	63	20	0
Bldg.S	402	2	1	12	541	41	0	229	3	167	118	8	340	275	153	14
BH	116	18	0	0	360	4	0	82	0	43	55	0	1	85	0	1
Kitchen	332	0	0	16	1199	130	23	256	19	117	180	76	1218	634	211	5

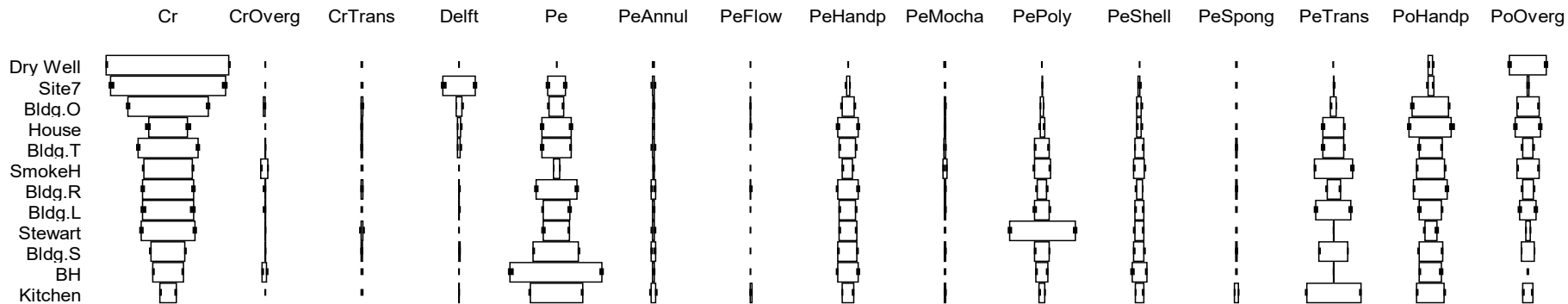
Site	MC Date
Dry Well	1781.4
Site7	1782.9
Bldg.O	1788.1
House	1793.2
Bldg.T	1794.9
SmokeH	1795.3
Bldg.R	1796.7
Bldg.L	1797.3
Stewart	1798.1
Bldg.S	1798.9
BH	1799.6
Kitchen	1804.3

Problems with the MCD and Pipe stem Formulas?

Problems with the MCD and Pipestem Formulas

- they are formulaic!
 - no matter what the data look like, you can always get an answer.
 - no indication if the solution fits the seriation model!

The Solution



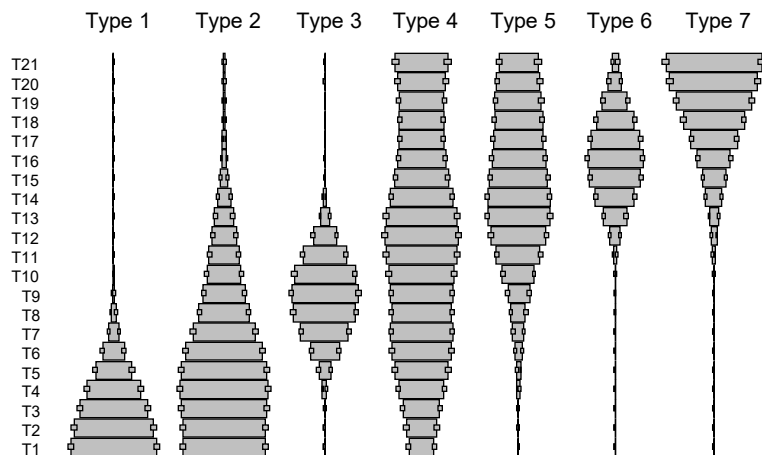
- Sort the assemblages in their MCDs and plot the resulting seriation diagram: do you see the battleship-shaped curves?
- Do two seriations and compare the results: do they agree?
 - pipestem bore diameter seriation order vs. ceramic seriation order
 - ratio of wrought /cut nails vs. ceramics seriation order

Correspondence Analysis

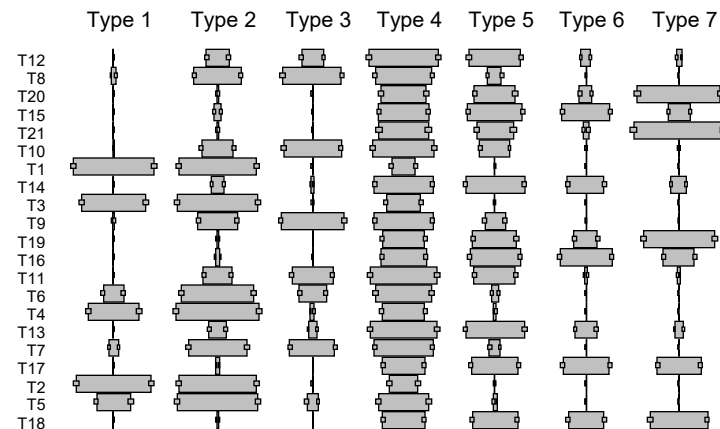
1. Iteration 1					
Type:	Creamware	Delft	Pearlware	WhiteSaltGlaze	Scores
Random Numbers:	0.273366978	0.760887	0.92061152	0.164707505	
Dry Well	0.80	0.00	0.00	0.20	0.251375
Site7	0.64	0.17	0.10	0.09	0.412147
Bldg.O	0.75	0.06	0.14	0.05	0.384773
House	0.55	0.04	0.40	0.01	0.552651
Bldg.T	0.62	0.04	0.29	0.05	0.47825
2. Iteration 2	1.348169562	0.137663	0.45538671	0.137976796	
Dry Well	0.80	0.00	0.00	0.20	
Site7	0.64	0.17	0.10	0.09	
Bldg.O	0.75	0.06	0.14	0.05	
House	0.55	0.04	0.40	0.01	
Bldg.T	0.62	0.04	0.29	0.05	
3. Iteration 3					
Dry Well	0.80	0.00	0.00	0.20	1.103232
Site7	0.64	0.17	0.10	0.09	0.942482
Bldg.O	0.75	0.06	0.14	0.05	1.087218
House	0.55	0.04	0.40	0.01	0.925372
Bldg.T	0.62	0.04	0.29	0.05	0.977357

Correspondence Analysis

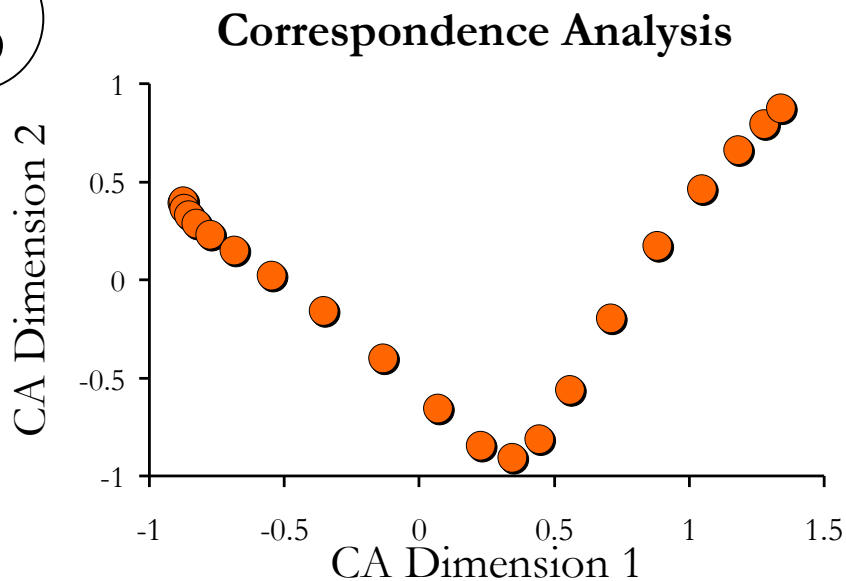
1



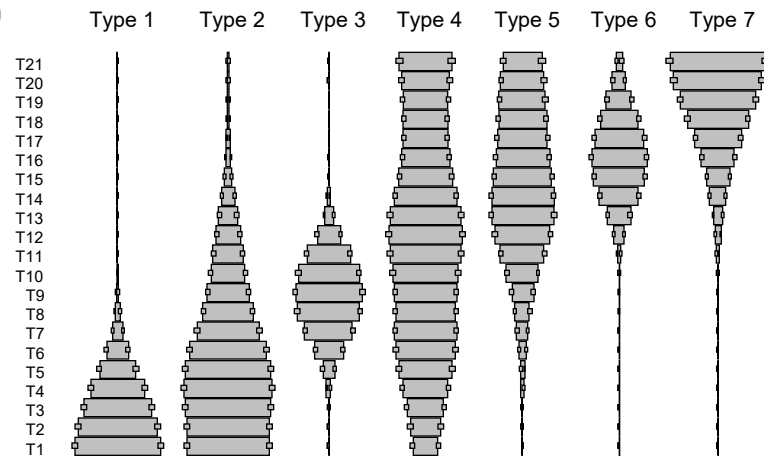
2



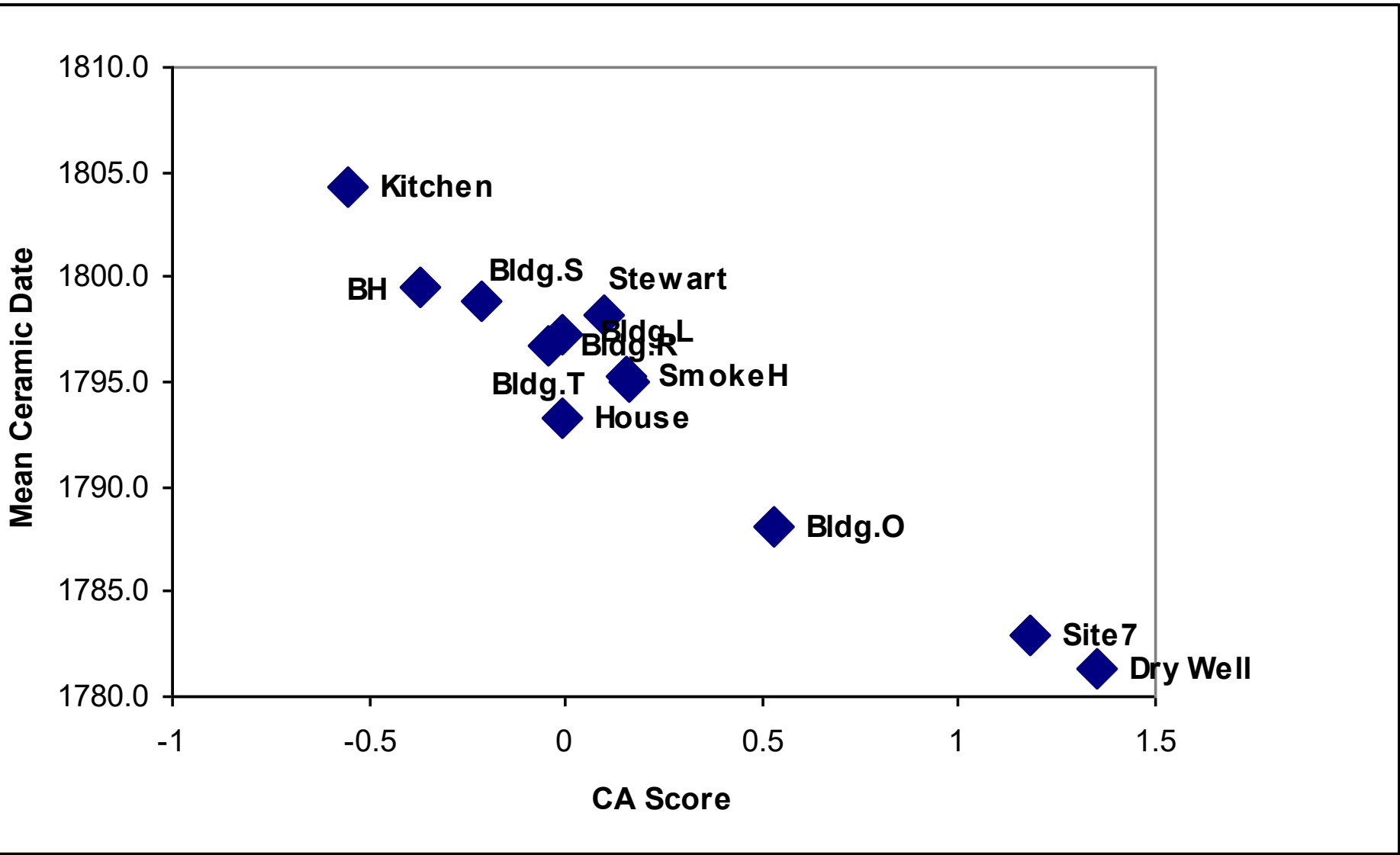
3



4



MCD vs. CA



For more on MCDs, CA, and more in R.....

<https://github.com/DAACS-Research-Consortium/DAACS-Open-Academy>

<https://www.daacs.org/research/playlists/doa-archives/doa-fall-2021/>

Frequency Seriation Methods

Geeky Comparison of Model Assumptions

1. Frequency Seriation

- types display battleship-shaped curves on a single gradient (time)
- temporal overlap of types
- same “local area” and “cultural tradition”
- similar time averaging across assemblages
- little sampling error

2. Correspondence Analysis

- all of the above. Plus....
- types display Gaussian curves on one or more gradients
- uniformly distributed type maxima, type variances, assemblages.

3. Mean Ceramic Dating

- all of the above. Plus....
- types display Gaussian curves on a single gradient
- type variances similar
- dates of type maxima are known

4. Pipe-stem Dating

- bore diameter class maxima are evenly spaced in time.